

Two rewrites in one branch.

- **Product:** vanilla HTML and scripts → React with a governed design system
- **Harness:** nothing pushed back → machines enforce the rules
- The second rewrite is what keeps the first one true
- Every claim in this deck carries a command you can run yourself

What is coming needs a frontend that can hold.

- A re-brand is ahead
- The interface needs consistent control across every surface
- Agents build user interface here daily
- On main, each of those means hand-editing pages that carry their own copy of the truth

Main is ungoverned — not ugly.

0 component states on record

- Credit where due: the palette is already centralized in 1 CSS file
- 2 pages still compile their styling in the browser at page load
- 400 test files — all behavior, none can see pixels
- The design system is copy-the-nearest-example. Drift is not that loop failing — drift is its output

receipts: R1 — press ↓ for the detail slides

Main already did the easy half.

1 CSS file holds the whole palette

- Centralizing color was done — credit where due
- 2 pages still compile their styling in the browser, on every load

```
node frontend/scripts/pr-case-tools.mjs --probe R1 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

What consistency costs on main — today.

1,979 hand-paired dark: variants

- 5,237 raw color classes across the product — 137 distinct values, every one a hand-made decision
- Dark mode is maintained by hand in 33 files — every light change needs its dark twin
- A re-brand means finding and editing all of them, and hoping nothing was missed
- Measured live at the pinned base every time this deck builds — press ↓ for the commands

Appearance has one source of truth.

1 file: tokens.json

- A valid edit propagates to the entire application
- An invalid edit is refused — 7 classes of broken edit, each with a plain-language error
- A re-brand stops being archaeology and becomes an edit plus a review

receipts: R2 — press ↓ for the detail slides

The guard says no — and says why.

7 classes of broken edit refused

- A valid token edit propagates to the whole application
- An inverted Canvas-to-Paper ladder is refused with a plain-language error
- Runs in a temporary directory — canonical files stay untouched

```
node frontend/scripts/pr-case-tools.mjs --probe R2 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

A temporary-directory token generation propagates one valid edit and refuses an inverted Canvas-to-Paper ladder. — re-derived by the command above; a mismatch fails check:design-system

ACCEPTED EDIT

```
semantic.light.primary -> oklch(0.32 0.08 255)
```

```
--primary: oklch(0.32 0.08 255);
```

propagates to the entire application

REFUSED EDIT

```
semantic.light.paper -> oklch(0.94 0 0)
```

```
Error: semantic.light Canvas to Paper lightness  
step -0.010 is below 0.020
```

the guard explains itself — no silent failure

values extracted from the checksummed case at build time — run R2 to re-derive both

Appearance is testable now — and nothing was lost.

5,503 root tests pass

- All 400 of main's test files kept, 3 added
- 99 story files render real component states
- A 28-stage gate must pass at one clean revision before anything ships

receipts: R3 · R4 — press ↓ for the detail slides

Every test main had is still here.

5,503 root tests pass

- 400 files kept, 0 removed
- 3 new files added on top

```
node frontend/scripts/pr-case-tools.mjs --probe R3 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

Component states are on record now.

99 story files render real states

- Backed by a 28-stage receipt-grade gate
- Before this branch, the number was zero

```
node frontend/scripts/pr-case-tools.mjs --probe R4 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

The branch carries 99 Storybook story files and a 28-stage receipt-grade gate. — re-derived by the command above; a mismatch fails check:design-system

A surface grammar, not a pile of screens.

- **Canvas** holds the page · one **Paper** holds the work · **Containers** group · one **Overlay** floats
- 77 catalogued components across 3 tiers — queryable, not memorized
- Humans and agents compose from a vocabulary instead of inventing layouts

receipts: R5 — press ↓ for the detail slides

The inventory is queryable.

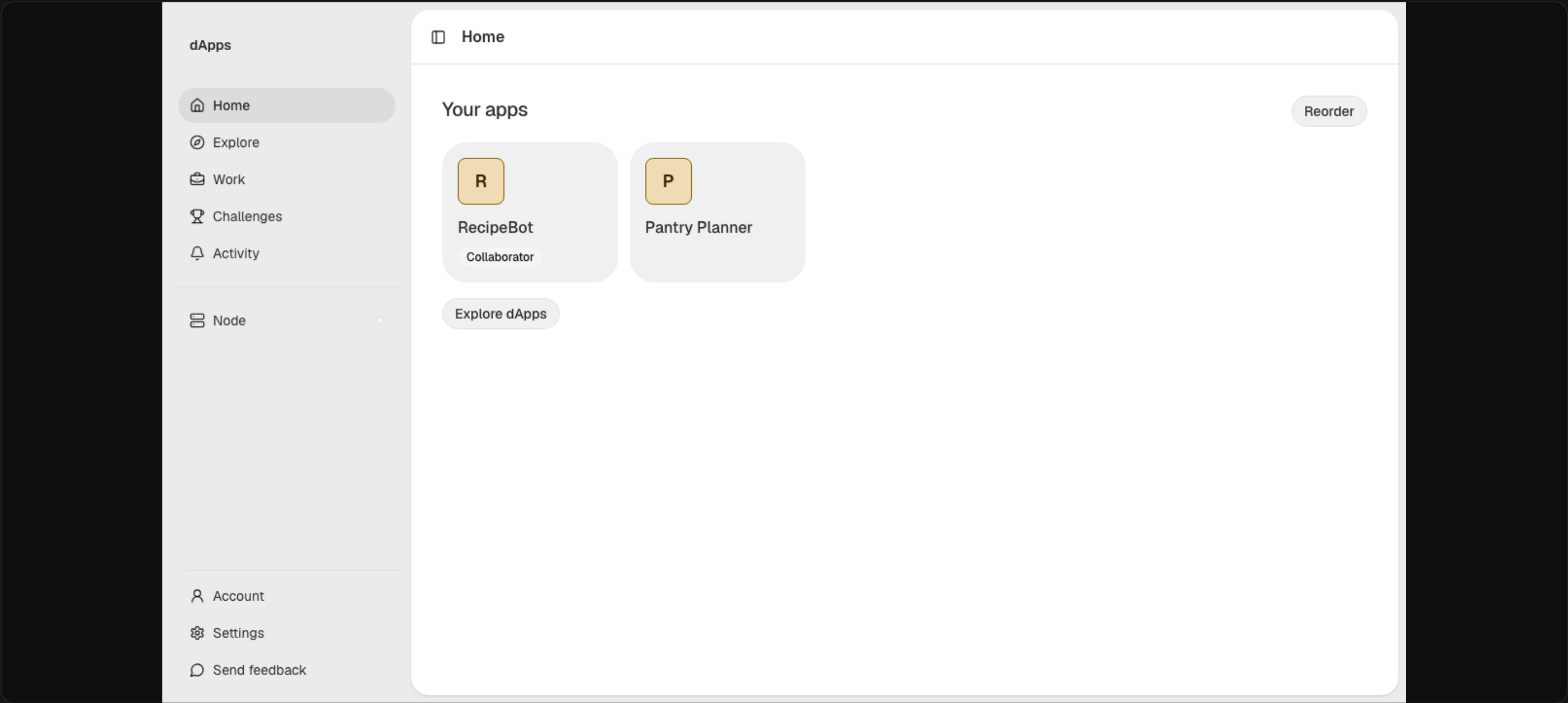
77 catalogued components

- 34 element · 28 block · rest feature — 3 explicit tiers
- Agents query the catalog; nobody bulk-reads a source directory

```
node frontend/scripts/pr-case-tools.mjs --probe R5 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

77 catalog entries expose the component inventory across 3 explicit tiers. — re-derived by the command above; a mismatch fails check:design-system

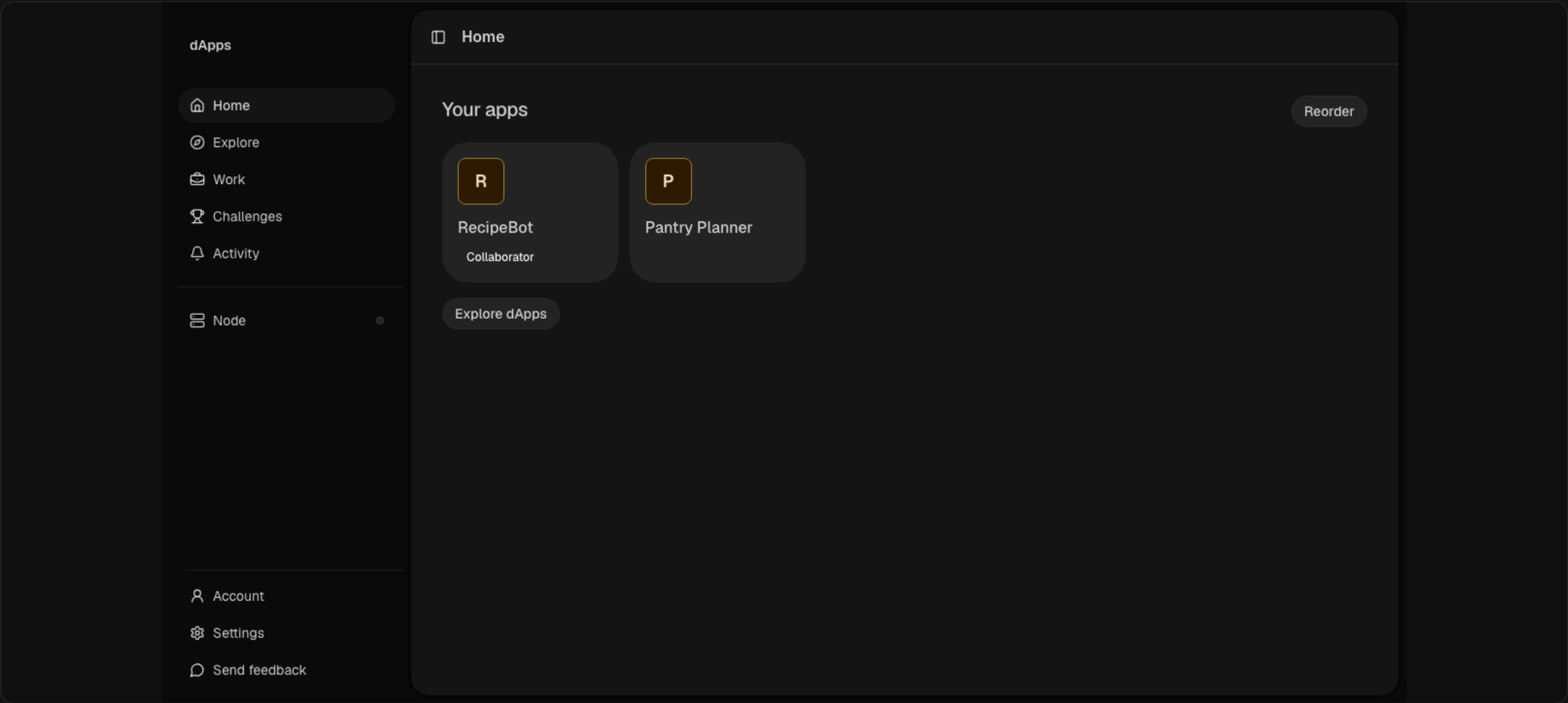
Home route at the pinned branch head.



Branch home route, light theme

click any image for full screen · Esc closes · ↑ returns

Same tokens re-resolved — no hand-paired dark variants anywhere.



Branch home route, dark theme

click any image for full screen · Esc closes · ↑ returns

The harness that was there before.

- **Guidance** was the neighboring code — copy the nearest example and adjust
- **Verification** was 400 behavior test files — kept and credited, but none can see pixels
- **Feedback** was save, reload, eyeball
- Drift was not that loop failing. Drift was its output

receipts: R6 — press ↓ for the detail slides

Coverage that could not see.

0 component states on record

- 400 behavior test files — kept and credited
- Not one of them renders a pixel

```
node frontend/scripts/pr-case-tools.mjs --probe R6 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

Pinned main has 400 root Node test files and zero component-isolated Storybook files. — re-derived by the command above; a mismatch fails check:design-system

Lean guidance in, mechanical proof out.

28/28 gate stages, zero skipped

- **Before generation:** a 17,378-byte law always loaded; the catalog is queried, never bulk-read
- **After generation:** a per-edit check that teaches, a gate that blocks
- It caught itself: guidance grew to 22,445 bytes against a 17,427-byte ratchet — refused, 3 failures preserved

receipts: R7 · R8 · R9 — press ↓ for the detail slides

Guidance stays small by law.

17,378 bytes always loaded

- Held under a 17,427-byte ratchet that only tightens
- Full component review: 21,257 bytes under a 21,268 ratchet
- The 100,000-byte ceiling is non-waivable — owner decision

```
node frontend/scripts/pr-case-tools.mjs --probe R7 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

Component-review context is 21257 bytes under a 21268-byte ratchet and 100000-byte owner ceiling. — re-derived by the command above; a mismatch fails check:design-system

One clean revision. Zero excuses.

28/28 gate stages passed

- 0 skipped, 0 omitted
- Start revision = end revision, working tree clean at both ends

```
node frontend/scripts/pr-case-tools.mjs --probe R8 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

28 of 28 gate stages passed at one clean immutable source revision. — re-derived by the command above; a mismatch fails check:design-system

The loop caught itself.

3 failures preserved as evidence

- Guidance grew to 22,445 bytes against a 17,427 ratchet — the gate refused
- The fix routed 5,018 bytes on demand. The ratchet did not move

```
node frontend/scripts/pr-case-tools.mjs --probe R9 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

This is not free. That is the point.

- 19 check scripts and a 28-stage gate are maintained infrastructure
- The trade: recurring review arguments become reproducible failures
- A command that exits 1 ends a debate that comments never could

receipts: R10 — press ↓ for the detail slides

The price tag, itemized.

19 check scripts to maintain

- Plus a 28-stage gate to keep green on every slice
- In exchange: 1 token source rules them all

```
node frontend/scripts/pr-case-tools.mjs --probe R10 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```


Review the constraint system, not 50,000 lines.

- Check the laws, the tokens, the guards — then spot-check that violations actually fail
- 35 of 159 commits carry provenance to 25 recorded decisions
- See it with your eyes: 4 pinned captures — same route, same data, both themes

receipts: R11 · R12 — press ↓ for the detail slides

Decisions have receipts.

35 of 159 commits carry provenance

- Each points at one of 25 recorded decisions
- The grammar arrived mid-program — earlier history is summarized, never backfilled

```
node frontend/scripts/pr-case-tools.mjs --probe R11 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

Same route. Same data. Both themes.

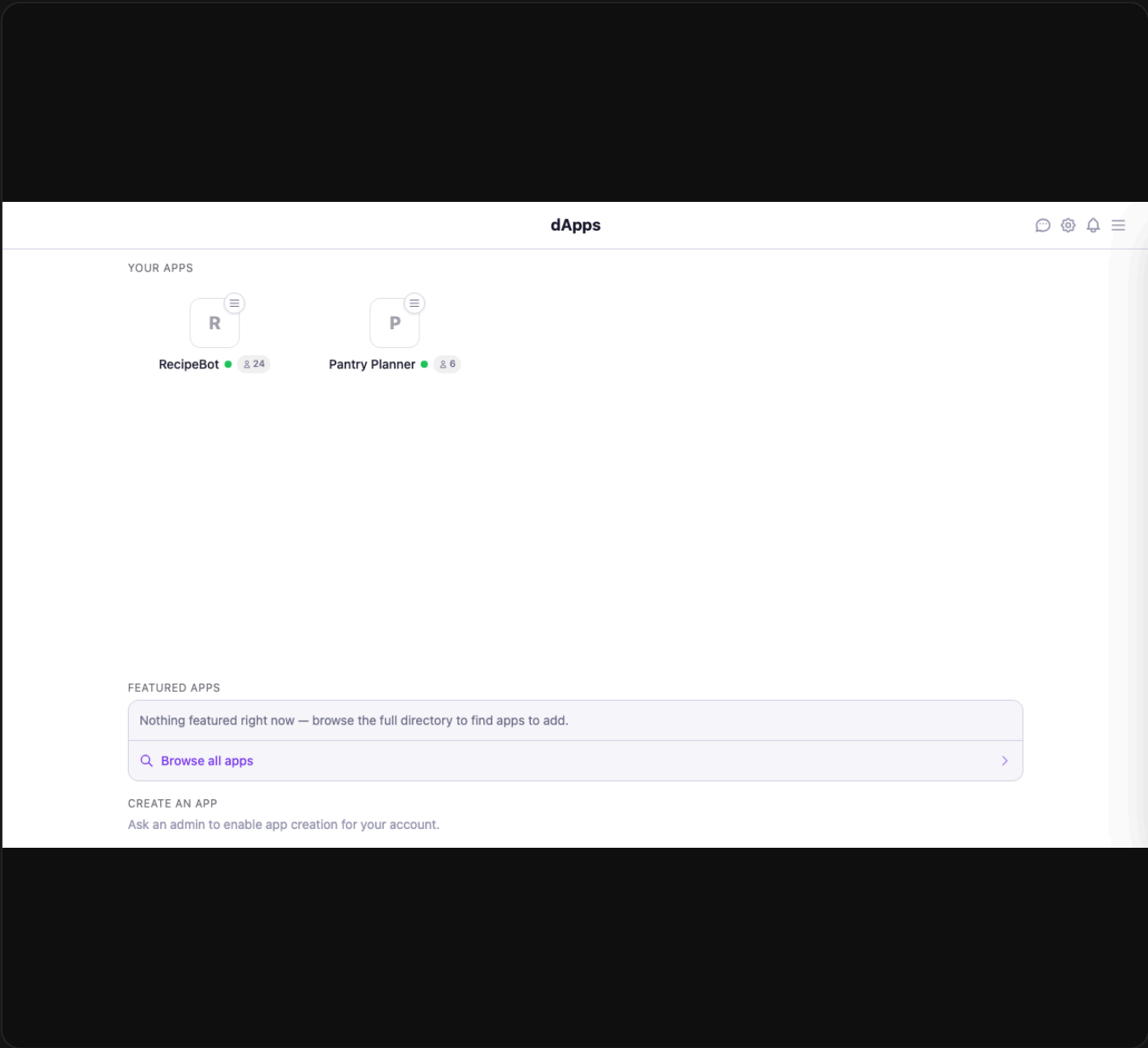
4 pinned captures

- 1280×720, one deterministic fixture, 2 themes
- Main and branch, side by side — the comparison is repeatable, not curated

```
node frontend/scripts/pr-case-tools.mjs --probe R12 --base 472de79151060300df68fc6e98e242351d76eef0 --branch 2f8605304ba9812474b2a8d8a8a0ff3fc3a83bf9
```

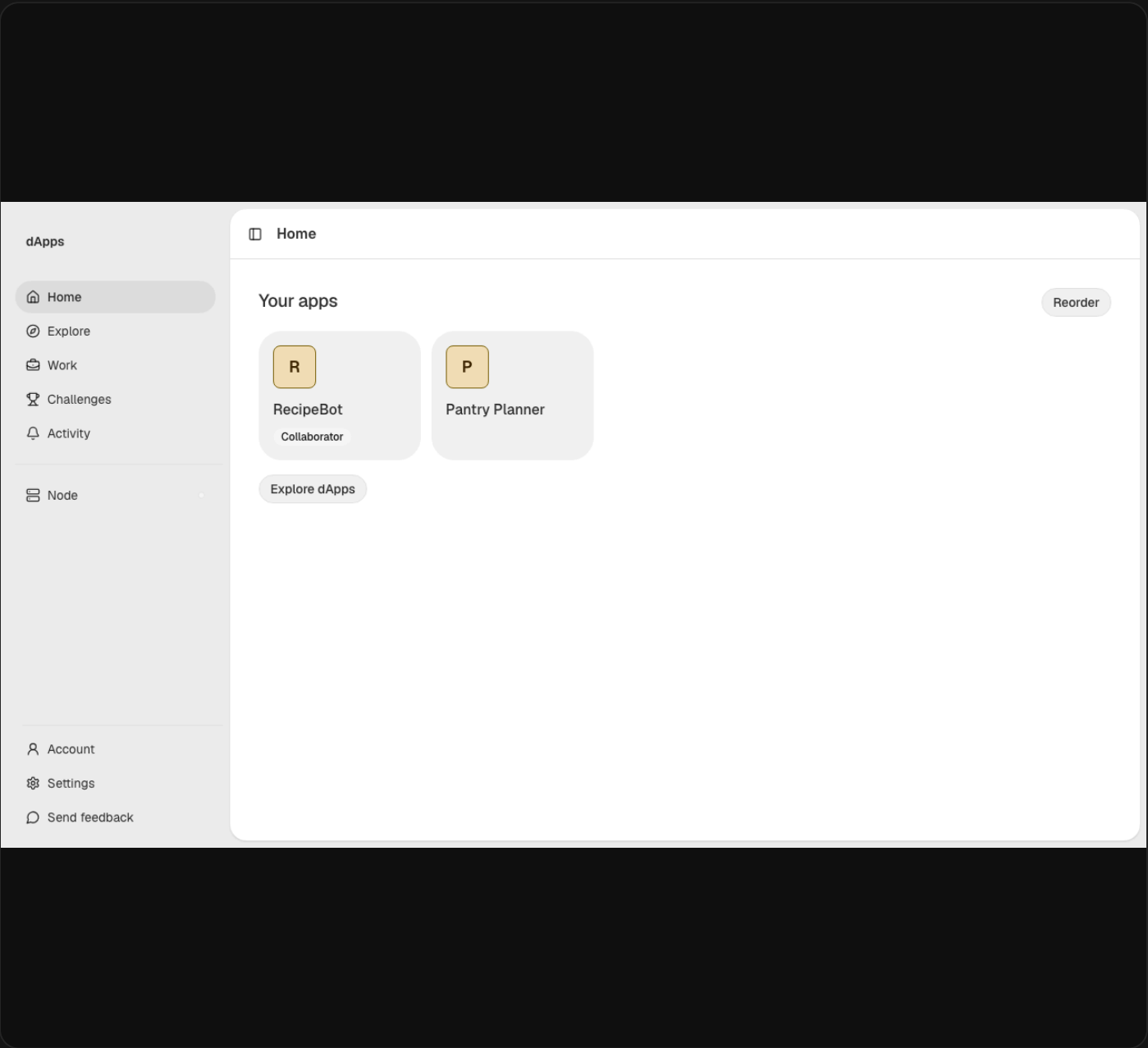
Four pinned Home captures compare current main and the integrated branch under one deterministic data fixture in light and dark themes. — re-derived by the command above; a mismatch fails check:design-system

Left: pinned main. Right: integrated branch. Same route, same fixture.



Main home route, light

click any image for full screen · Esc closes · ↑ returns



Branch home route, light

Three independent answers, all yours to run.

- **The machine:** a 28-stage gate at one clean, immutable revision
- **The suite:** 5,503 root tests, including every one main had
- **The reader:** twelve R-commands re-derive every number in this deck against committed claims
- If any of the three disagrees with a slide, the slide is wrong — that is the contract

Everything deeper lives in Storybook.

- Foundation pages, decision ledger, component index — generated into Storybook
- Same loop as the code: generated from source, citation-checked, never hand-typed
- Run `cd frontend && npm run storybook`
- Press ↓ for all twelve reproduce commands

